Tech Guardians

Vulnerability Assessment & Penetration Testing

USER GUIDE

Step-by-step: install, log in, scan and report

Version 2.0 · October 2026

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1. Before you start

You need:

- A Windows, Linux or macOS computer.
- **Python 3.9+** installed (tick *Add Python to PATH* on Windows).
- Internet access the first time (to install dependencies) and to scan targets.
- **Written authorisation** to test any target.

The tool is portable and needs no database — you can run it from a USB stick. Reports are saved in the **reports/** folder.

2. Install & launch the portal

1 Copy the folder

Unzip the tool anywhere — e.g. your USB stick or Desktop.

2 Run the launcher

Windows: double-click **run.bat**.

Linux / macOS: open a terminal in the folder and run:

```
chmod +x run.sh
./run.sh
```

3 Wait for first-time setup

The first launch creates a local environment and installs dependencies (one minute, needs internet). Later launches are instant.

4 Open the portal

Your browser opens automatically at:

```
http://127.0.0.1:5000/
```

If not, type that address into your browser.

3. Set a login password (securing access)

By default the portal opens without a password. To require a login so only you can use it, do this once:

1 Create a password hash

Open a terminal in the tool folder and run (replace *MyStrongPass*):

```
python -c "import hashlib;print(hashlib.sha256(b'MyStrongPass').hexdigest())"
```

Copy the long line it prints.

2 Edit config.py

Open **config.py** and set these values:

```
LOGIN_REQUIRED      = True
LOGIN_USERNAME      = "admin"
LOGIN_PASSWORD_HASH = ""
SESSION_SECRET      = "any-long-random-text"
```

3 Restart the portal

Close and re-launch. The portal now shows a sign-in page. Your password is stored only as a hash, never in plain text.

Tip — to change the password later, repeat step 1 with a new password and replace the hash. To turn login off again, set **LOGIN_REQUIRED = False**.

4. Log in

1 Enter credentials

On the sign-in page, type the username (default **admin**) and your password, then click **Sign in**.

2 You're in

You land on the assessment page. A **Log out** button appears top-right; use it when you finish.

Forgot the password? You can't recover a hash — just set a new one (section 3).

5. Run your first assessment

1 Enter the target

Type the website or host, e.g. **example.com** or **https://example.com**.

2 Fill engagement details (optional)

Client / Organisation, Assessor and Engagement Ref appear on the report cover.

3 Choose scan options

Tick which to include: installed open-source/Kali tools, the built-in port scan, and whether to allow **active** tools (more thorough, more intrusive).

4 Confirm authorisation

Tick the box confirming you have written permission to test the target. **The scan will not start otherwise.**

5 Start

Click **Start Assessment**. A live progress page shows each step; it moves to the results automatically when finished.

6. Read & download the report

The results page shows the overall risk rating, a security grade (A–F), a severity breakdown and every finding with remediation advice.

- **Download PDF** — the full branded report.
- **JSON** — machine-readable data.
- Both are also saved automatically in the **reports/** folder.

The PDF contains: cover, document control, executive summary with charts, scope & methodology, findings summary table, detailed findings (with CVSS, CWE, OWASP mapping, evidence and remediation), a remediation roadmap, OWASP compliance mapping and appendices.

7. Lock the tool to your machines (licensing)

Optional: bind the tool so it only runs on machines you authorise, using signed licenses. You keep a secret key; nobody can make a valid license without it.

1 Create your signing keys (once)

```
python license_tool.py genkeys
```

Keep **rjhex_private.key** secret. Paste the printed public key into **config.py** → LICENSE_PUBLIC_KEY, and set **LICENSE_ENFORCE = True**.

2 Get the user's machine code

On the machine that will run the tool:

```
python license_tool.py fingerprint
```

3 Issue a license

```
python license_tool.py issue --licensee "Acme" \
   --fingerprint  --days 365 \
   --private rjhex_private.key --out license.key
```

Send **license.key** to the user; they place it next to app.py. It works only on that machine, until it expires.

Full details, and how to compile to a single .exe for stronger protection, are in **SECURITY.md**.

8. Rebrand (name & logo)

- Edit **config.py** — change BRAND_NAME, tagline, colours and report settings.
- Replace the logo in BOTH **static/logo.svg** and **report/assets/logo.svg** (same file).
- Use an **SVG with solid colours** (no gradients) so it renders in the PDF.
- Match **static/style.css** *:root* colours to your config colours.

The author credit at the bottom of every page is set by AUTHOR_MARK in config.py.

9. Install extra VAPT tools

The portal auto-detects installed open-source / Kali tools and uses whatever is present (nmap, nikto, nuclei, wpscan, whatweb, wafw00f, sslscan, testssl.sh, subfinder, gobuster, ffuf and more). Example on Kali:

```
sudo apt update && sudo apt install -y nmap nikto whatweb \
  wafw00f dnsrecon sslscan testssl.sh dirb wapiti amass wpscan
```

The full install list (Linux/macOS/Windows) is in **TOOLS.md**. Installed tools show a green ✓ on the home page.

10. Troubleshooting & FAQ

The browser didn't open	Go to http://127.0.0.1:5000/ manually. If the port is busy, set TG_PORT to another value before launching.
"Python was not found"	Install Python 3 and tick *Add to PATH* (Windows), then re-run the launcher.
A scan shows errors / few findings	The target may block requests, or you're offline. Errors are listed in the report appendix; built-in checks still run even if external tools are absent.
I forgot my login password	Passwords are stored as a hash and can't be recovered — set a new one (section 3).
"LICENSE CHECK FAILED"	The machine isn't licensed or the license expired. Run the fingerprint command and request a new license.
Logo missing in the PDF	Use an SVG with solid fills — gradients don't render in the PDF.

11. Legal & authorised use

This tool performs vulnerability assessment and authorised penetration testing. Scan only systems you own or have **written permission** to test. Unauthorised scanning is illegal in most jurisdictions. You are responsible for holding authorisation and complying with all applicable laws, regulations and standards (e.g. CERT-In guidance).

Note on certification: CERT-In empanelment is granted to auditing organisations, not to software. This tool produces reports at a professional standard but cannot by itself make anyone CERT-In certified.